CARDIOVASCULAR PREDICTIVE DIGITAL TWIN

AURA PulseTwin

Proactive 7-Day Pre-Decompensation Forecasting & Clinical Decision Support for Post-MI Recovery

POST-MI DAYS 1–30 • P10 / P50 / P90
PURE-NUMPY CAUSAL PIPELINE • LOCAL SANDBOX



Proof-of-Concept Digital Twin in Healthcare
AURA BioEngineering Labs · MIT License
Institution: Jaipur Engineering College and Research Centre
Team: CIPHER 2.0
-Piyush Garg[Lead]
· Prakhar Vyas[Member 2]
· Pranav Katara[Member 3]
Github repo: https://github.com/Piyush-Garg-06/CIPHER-2.0_JECRC



The Post-Discharge Blindspot

Why Indian hearts fail too early — after the hospital's visibility ends

< 50

50%+ of Indian heart attacks strike under 50

30 DAYS

Critical gap from discharge to review

3× SIGNALS

HRV + non-dipping pulse drift 5–10 days early

POST-DISCHARGE VOID
Zero physiological visibility

SUB-CLINICAL DEGRADATION
Patients feel fine as reserves fail

ALARM FATIGUE PARADOX
Noisy ectopics get ignored

Convert passive wearables into actionable clinical foresight — with zero alarm fatigue.

From Reactive Thresholds to a Causal Digital Twin

The signal is latent physiological drift from a personal baseline — not a single vital

STATIC EHR
LVEF · HbA1c · eGFR
Comorbidity · prior PCI

DYNAMIC IoT
Resting HR · PPG HRV
Steps · sleep rhythm

CAUSAL TIME
Day t uses Day ≤ t
Zero future leakage

REACTIVE CARE
Day 0 · Emergency ICU
Mortality high · ₹5L+

AURA DIGITAL TWIN
Day −7 · Outpatient tele-rescue
Lower mortality · minimal cost

Production-Grade Full-Stack Architecture

Traceable flow from wearable Bluetooth sensors to bedside decision support

01 INGESTION

BLE 5.3 wearables
HR · HRV · SpO_2 · steps · sleep

02 API GATEWAY

FastAPI lifespan manager
SSE telemetry hub

03 PREDICTIVE CORE

Baseline lock · Kalman filter
Calibrated forecast · hysteresis

04 LOCAL STORAGE

SQLite 3 WAL · loopback only
No cloud PHI transmission

05 DECISION SURFACES

Ward matrix · 3D twin
XAI · What-If simulator

REACT 19 · PYTHON 3.12 · FASTAPI · SQLITE WAL · PURE-NUMPY · 56 PYTESTS / 100% PASS · DPDP-READY

Zero Black-Box. 100% Causal.

Auditable mathematics replacing unexplainable deep-learning hallucinations

01

BASELINE GUARD
Days 0–14 calibration
≥3 valid days

02

2-STATE KALMAN
Latent drift d_t + velocity
under measurement noise

03

CALIBRATED FORECAST
15 causal features + Platt
P10 · P50 · P90

04

TRIAGE STATE MACHINE
2-night hysteresis
72h episode chaining

$$x_t = A \cdot x_{t-1} + w_t \quad | \quad \text{Logit} = w^T z_t + b \quad | \quad P(\text{Decomp} \le 7d) = 1 / (1 + \exp(-(A_p \cdot \text{Logit} + B_p)))$$

Hospital Ward 8-Bed Live Triage Matrix

One surface for duty cardiologists and step-down ward nurses

SSE STREAM ACTIVE · sub-second

ALL 8 RED 2 YELLOW 3 WATCH 1 STABLE 2

Aarav Mehta · BED 01

HR 68 BP 130/80 SpO_2 96%
RISK 12%

Eleanor Vance · BED 02

HR 71 BP 131/81 SpO_2 97%
RISK 84%

Rahul Sharma · BED 03

HR 74 BP 132/82 SpO_2 98%
RISK 24%

Meera Iyer · BED 04

HR 77 BP 133/83 SpO_2 96%
RISK 61%

Kabir Singh · BED 05

HR 80 BP 134/80 SpO_2 97%
RISK 7%

Nisha Rao · BED 06

HR 83 BP 130/81 SpO_2 98%
RISK 46%

Dev Patel · BED 07

HR 86 BP 131/82 SpO_2 96%
RISK 76%

Anita Bose · BED 08

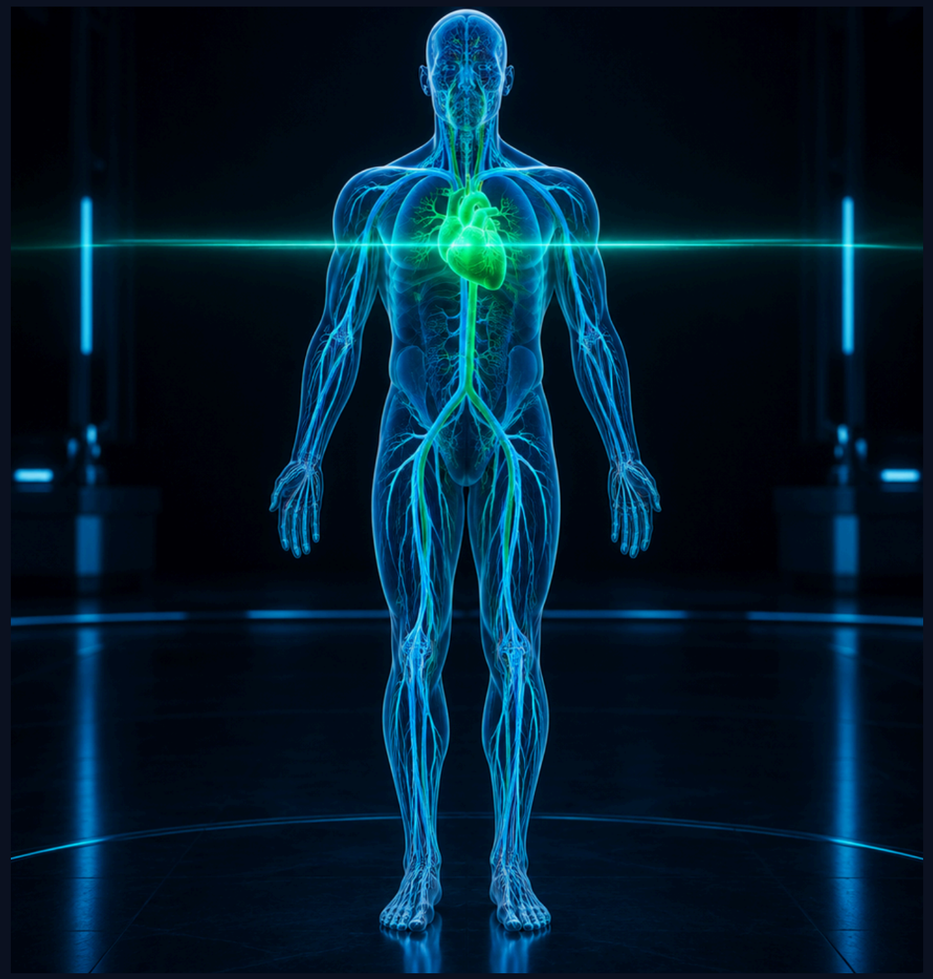
HR 89 BP 132/83 SpO_2 97%
RISK 18%

Patient Personal Health Twin

A living physiological replica — not a static avatar

RESTING HR
82 bpm
+8.4% · HR_0 74

ARTERIAL BP
138/88
Stage 1 hypertension



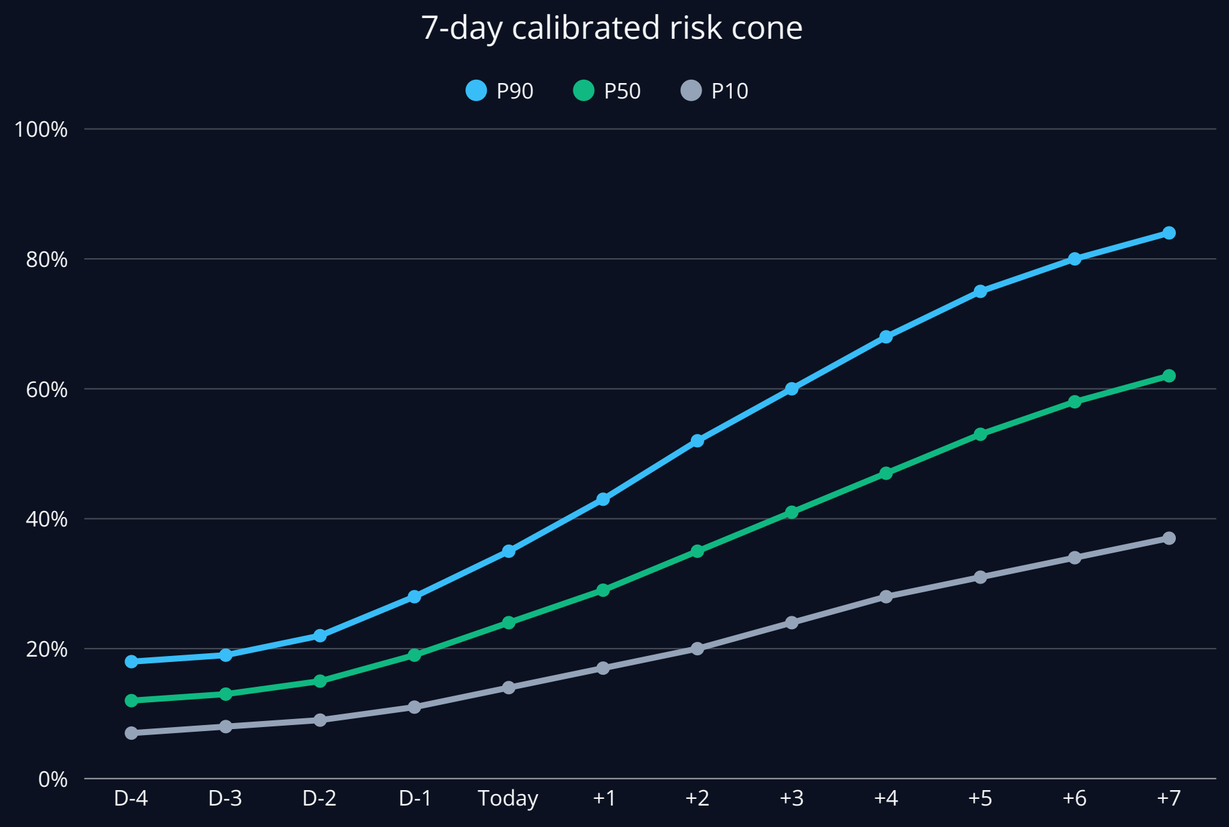
OXYGEN
97% SpO_2
Normal sinus status

HRV RECOVERY
42 ms
rMSSD autonomic trend

PHYSIOLOGICAL LOAD REST STRESS

Seven Days of Transparent Foresight

Quantified confidence cones and exact log-odds feature attribution



EXACT LOG-ODDS DRIVERS

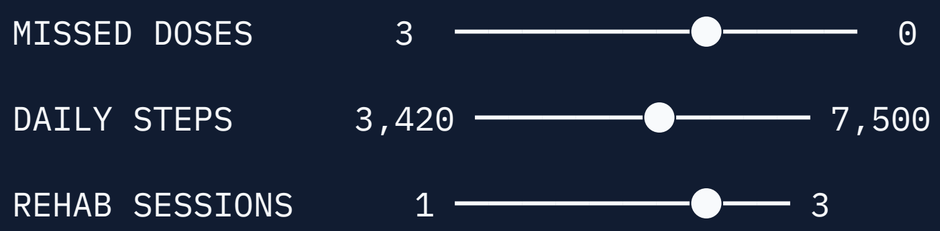
Recovery Index (d_t)	+0.82
Resting HR Elevation	+0.41
Kinetic Mobility Deficit	+0.34
Non-smoker	−0.21
Cardiac Rehab attendance	−0.15

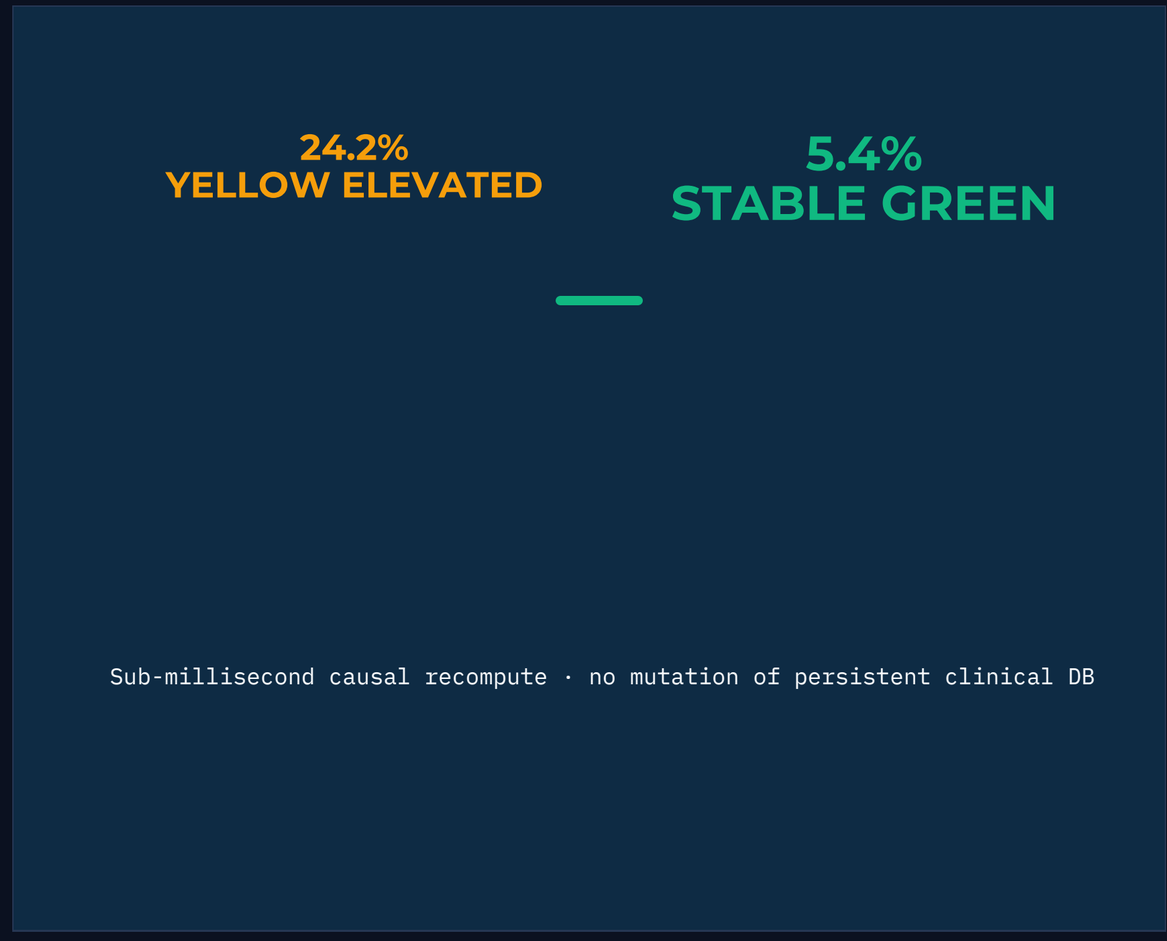
Prescription by Simulation

Interactive counterfactual reasoning for shared clinical decision-making

RAHUL SHARMA

Eliminate missed medication and achieve the daily movement goal.





Triage in Sync. Records in Context.

Dual-role alert management paired with a FHIR-ready longitudinal archive

ALERT CENTER

ACTIVE ALERTS + CLINICAL RATIONALE

Episode V2-PT-8092-EP-D16
72-hour symptom deduplication

2-night temporal hysteresis
Alarm fatigue shield

LONGITUDINAL EHR VAULT

48h Holter ECG
High-Sensitivity Troponin-I
NT-proBNP biomarkers

Post-PCI Echo · LVEF 42%
Authenticated discharge PDF

Condition-Specific Education & Clinical AI Co-Pilot

Empowering post-MI patients and treating physicians alike

CARDIAC REHAB VIDEO HUB

ADULT CPR

10-MINUTE POST-MI HOME EXERCISES

STRESS & VAGAL RECOVERY

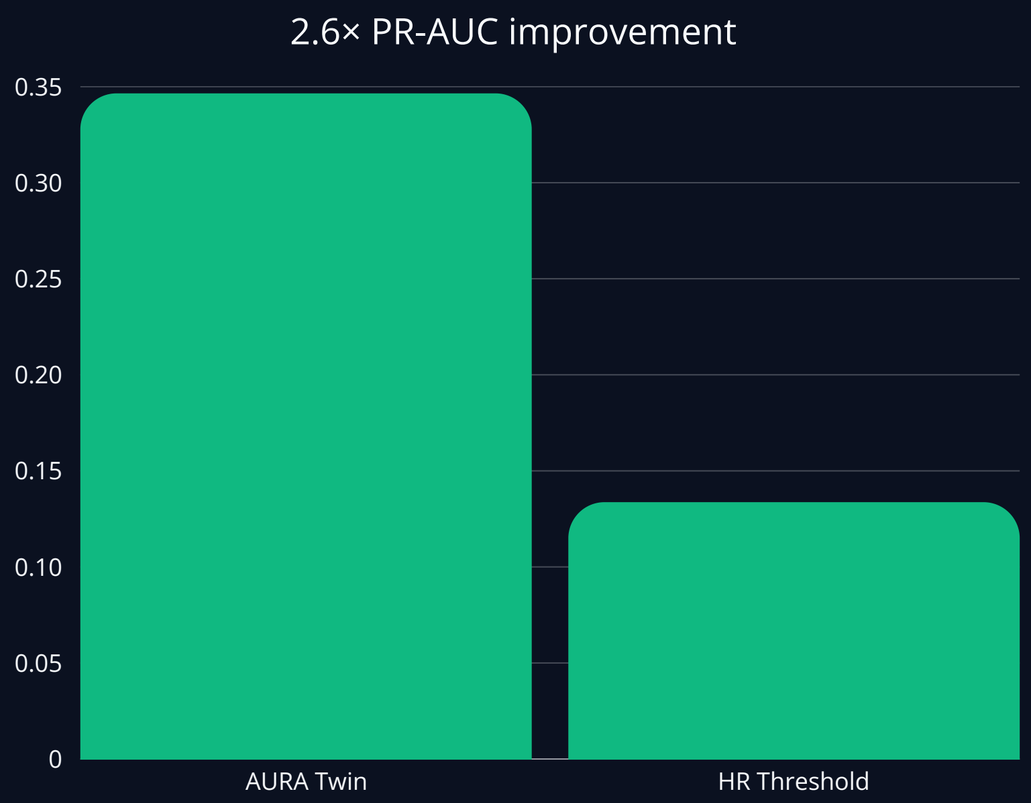
AURA CLINICAL COMPANION

CLINICIAN MODE
Guideline synthesis · contraindication checks · bedside escalation summaries

PATIENT MODE
Jargon-free, empathetic explanations of biometric shifts and next actions

Validated Clinical Superiority. Lean Computation.

2,000-patient synthetic cohort · 60/20/20 split · 1,000-resample bootstrap



70.9%
ROC-AUC

3.0 DAYS
Early warning lead

63%
False alarm reduction

56 / 56
PYTESTS PASS

Built for Foresight — Not Just Detection

AURA combines calibration, causal forecasting, explainability and local privacy

CAPABILITY	CONSUMER SMARTWATCH	TRADITIONAL RPM	BLACK-BOX DEEP LEARNING	AURA PULSETWIN
Physiological approach	Threshold	Threshold	Pattern match	Causal 4-stage twin
Baseline calibration	Limited	Partial	Black-box	Individualized days 0–14
Forecast horizon	None	Hours	7d uncalibrated	7d calibrated P10/P50/P90
Alarm fatigue shield	No	Limited	No	2-night + 72h dedup
Interpretability	Low	Medium	Uninterpretable	Exact log-odds XAI
Privacy	Cloud-dependent	Mixed	Heavy / opaque	Local WAL · DPDP-ready

Reforming Cardiovascular Recovery Across India

From validated prototype to ABDM-connected public health deployment

NOW · COMPLETED

56 tests · 4-stage engine
React 19 / FastAPI surfaces

MONTH 6

Tier-1 cardiology ward pilot
Step-down post-PCI patients

MONTHS 12–18

ABDM-connected public health deployment across India

EARLIER INTERVENTION · **LESS ALARM FATIGUE** · **LOCAL PRIVACY** · **SCALABLE RECOVERY**

AURA PulseTwin — foresight before failure.